

Responses Of Serum IGF-1 After An Acute Bout Of Lower-Body Resistance Exercise.

Bill I. Campbell, Paul La Bounty, Austin Oetken, Mike Greenwood, Richard Kreider, Darryn Willoughby

The acute response of IGF-1 to resistance exercise remains unclear. Most studies have shown no change in IGF-1 during or immediately following an acute bout of resistance exercise, whereas a few studies have shown acute elevations during and following resistance exercise. **PURPOSE:** To determine the serum free IGF-1 response to an acute bout of intense, lower body resistance exercise. **METHODS:** Ten healthy and physically active males (21 ± 2.4 yrs, 83 ± 11.2 kg, 176.8 ± 7.7 cm) engaged in an acute bout of lower body RE which consisted of four sets of both leg press and leg extension at 80% 1RM to failure. Rest periods between sets and exercises were approximately 150 seconds. Immediately prior to the resistance exercise bout, but following an 8-12 hour fast, participants underwent an initial/baseline blood draw. Blood was also obtained immediately, 30 minutes, two and six hours after the resistance exercise bout. Serum free-IGF-1 was analyzed via ELISA [Active® Bioactive ELISA by Diagnostic Systems Laboratories Inc. (DSL-10-9400; Webster, TX)]. A one-way ANOVA with repeated measures was utilized to analyze the data. Following the ANOVA, a series of paired-samples t-tests (one for each time point compared to baseline values) was conducted. **RESULTS:** Data are presented as means $\pm$ standard deviation utilizing ng/ml as the units of measurement. Baseline free IGF-1 = 1.2 ± 0.43 ng/ml. Following exercise, free IGF-1 concentrations were 1.6 ± 0.54 ; 1.5 ± 0.57 ; 1.4 ± 0.46 ; and 1.1 ± 0.53 ng/ml immediately post, 30 minutes, 2 hrs, and 6 hours post exercise, respectively. The ANOVA analysis indicated a significant difference across time for IGF-1 ($p < .001$). Subsequent paired-samples t-tests revealed a significant difference between the baseline and immediate post-resistance exercise IGF-1 levels ($p = .001$). Statistical trends were observed between baseline and 30 minutes post-exercise ($p = .053$) and baseline and 2 hours post-exercise ($p = .052$). **CONCLUSIONS:** An acute bout of lower-body resistance exercise in which each set is performed to failure significantly increases serum IGF-1 levels. The findings from the present study support those from other studies but are in contrast to others. Possible differences accounting for the discrepancies of the acute IGF-1 response to resistance exercise include training to failure on each set, rest periods between sets, exercise selection, or other variables. More investigations involving the response of IGF-1 to intense resistance exercise are needed to further elucidate the responses of this anabolic hormone. **PRACTICAL APPLICATIONS:** Resistance training improves muscular strength, muscular endurance, and increases lean body mass. Serum IGF-1, which increases in response to intense resistance exercise, may be (at least in part) responsible for some of these functional and physical adaptations in skeletal muscle.